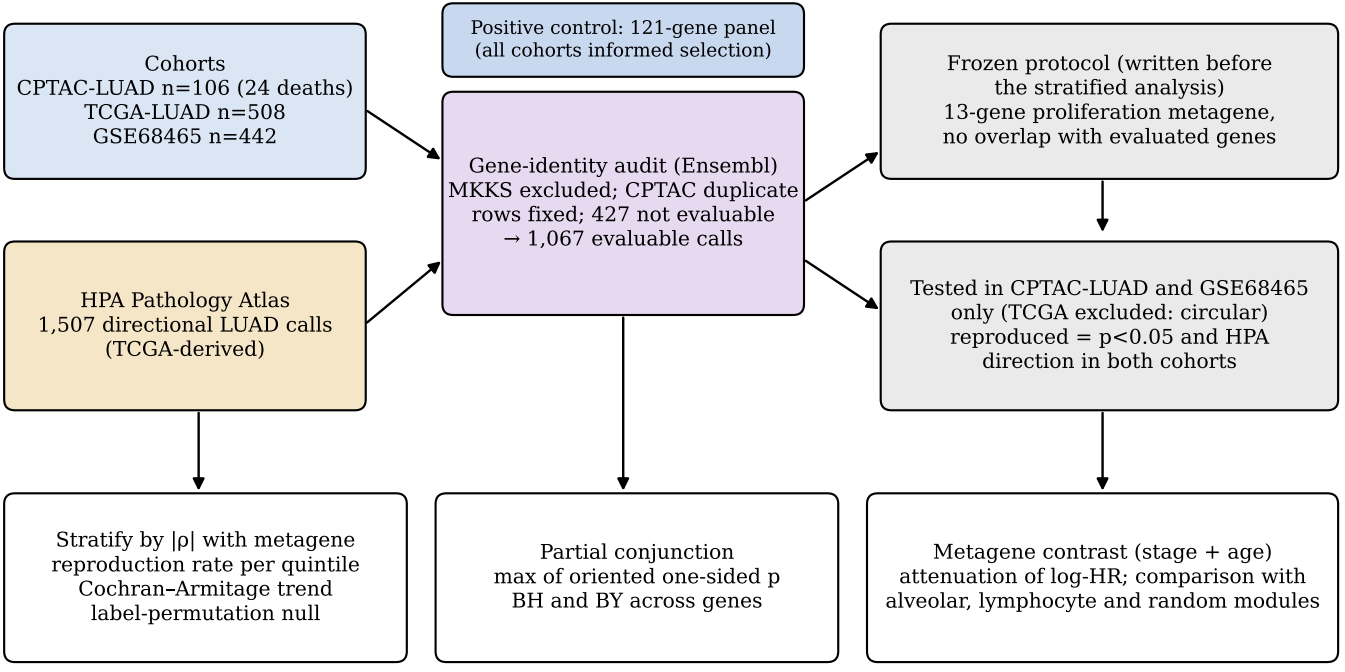
Cohorts
CPTAC-LUAD n=106 (24 deaths)
TCGA-LUAD n=508
GSE68465 n=442
Positive control: 121-gene panel
(all cohorts informed selection)
Frozen protocol (written before
the stratified analysis)
13-gene proliferation metagene,
no overlap with evaluated genes
Gene-identity audit (Ensembl)
MKKS excluded; CPTAC duplicate
rows fixed; 427 not evaluable
→ 1,067 evaluable calls
HPA Pathology Atlas
1,507 directional LUAD calls
(TCGA-derived)
Tested in CPTAC-LUAD and GSE68465
only (TCGA excluded: circular)
reproduced = p<0.05 and HPA
direction in both cohorts
Stratify by |ρ| with metagene
reproduction rate per quintile
Cochran–Armitage trend
label-permutation null
Partial conjunction
max of oriented one-sided p
BH and BY across genes
Metagene contrast (stage + age)
attenuation of log-HR; comparison with
alveolar, lymphocyte and random modules